The Bilal Morning

Tuesday, 13 October 2026 BACKFILLED EDITION 12 stories · 17 sources

1 source articles could not be read today; those stories show headlines and links only.

THREE THINGS TO KNOW

1. Pluto's retrograde has ended, opening an eight-month window for personal transformation and community ties.
2. NASA and the Department of Energy have partnered to advance nuclear space technologies.
3. Astronomers have discovered an atmosphere on the rocky exoplanet LHS 1140 b, boosting the possibility of liquid water.

Pluto Retrograde Ends as Planet of Transformation Resumes Forward Motion

As Pluto concludes its retrograde period, the zodiac enters an eight-month window to pursue personal transformation and strengthen community ties.

- Pluto's retrograde concludes on October 15, returning the planet of transformation to direct movement. [iowastatedaily.com]
  > “With Pluto, the planet of transformation, now moving forward again, you can start setting change into motion.”
- The zodiac will have an eight-month period to focus on goals before the next Pluto retrograde begins in May 2027. [iowastatedaily.com]
  > “The next time this retrograde will happen will be May 8, 2027, affording the Zodiac 8 patient months to work towards their goals.”
- During Pluto's forward transit, people are advised to foster close relationships and contribute to their communities. [iowastatedaily.com]
  > “the signs are urged to invest in their communities and strengthen close relationships across the board.”
- Pluto's stay in Aquarius will span a total of twenty years, lasting until 2044. [iowastatedaily.com]
  > “Pluto is and will be in Aquarius for the coming two decades (2024-2044).”

In the image: The image shows an inlaid geometric floor medallion featuring rings of the twelve astrological zodiac signs, illustrated with both pictorial figures and their corresponding glyph symbols.

✓ 4 of 4 claims verified against sources · 1 source

iowastatedaily.com

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ASTRONOMY AND ASTROPHYSICS

Pulsar Helium Outlines Financing Strategy for Minnesota Rare Gas Hub

Pulsar Helium aims to limit new equity by prioritizing debt and public-sector financing for its proposed Minnesota plant.

- Pulsar Helium accepted a binding proposal from Chart Energy & Chemicals for plant equipment totaling $85.5 million.
  [brantfordexpositor.ca]
  [elliotlakestandard.ca]
  > “accepted a binding proposal (the “Proposal”) from Chart Energy & Chemicals, Inc., a wholly owned subsidiary of Baker Hughes Company, establishing a staged framework for the supply of equipment for the Plant at an aggregate equipment-supply price of US$85.5 million.”
- The helium system features a nameplate liquefaction capacity of 861 liters per hour with a product purity of 99.999%.
  [brantfordexpositor.ca]
  [elliotlakestandard.ca]
  > “The helium system has a stated nameplate liquefaction capacity of 861 liters per hour, equivalent to approximately 7.5 million liters of liquid helium annually at continuous nameplate operation”
- The carbon dioxide system has a stated processing capacity of 300 metric tonnes per day and a purity of 99.99%. [brantfordexpositor.ca]
  [elliotlakestandard.ca]
  > “The CO2 system has a stated processing capacity of 300 metric tonnes per day, equivalent to approximately 109,500 metric tonnes annually at continuous nameplate operation”
- The company is pursuing parallel financing discussions with University Bank, DEED, USDA Rural Development, and IRRR.
  [brantfordexpositor.ca]
  [elliotlakestandard.ca]
  > “We are progressing multiple financing channels in parallel, including discussions with University Bank, DEED, USDA Rural Development and IRRR.”

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How Ancient Observers Used the Stars to Prove Earth Is Round

For thousands of years, observations of changing star elevations across different latitudes have demonstrated that Earth is a sphere.

- Eratosthenes used shadow lengths in Egypt to calculate Earth's circumference and prove it is spherical.
  [noozhawk.com]
  > “he used simple geometry to not only show that the Earth is spherical but also to calculate its circumference”
- Aristotle noticed that travelers going south saw stars rise higher above the horizon. [noozhawk.com]
  > “the farther south they traveled, the more the stars in that direction appeared higher in the sky.”
- Piscis Austrinus features Fomalhaut as the brightest and easiest star grouping to spot. [noozhawk.com]
  > “The highest, and easiest to spot, is the ancient constellation of Piscis Austrinus”
- Abbé Nicolas-Louis de Lacaille mapped southern constellations from the Cape of Good Hope. [noozhawk.com]
  > “who mapped the stars of the Southern Hemisphere from the Cape of Good Hope from 1751 through 1753.”

In the image: The image shows an astronomical diagram with coordinate lines illustrating the approximate northern latitude horizons and visible southern constellations.

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noozhawk.com

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First Potentially Habitable Rocky Exoplanet Found With an Atmosphere

Astronomers discovered an atmosphere on the rocky exoplanet LHS 1140 b, boosting the possibility of liquid water and advancing the search for extraterrestrial life.

- Researchers found the first atmosphere around a rocky exoplanet located in its star's habitable zone.

  [timesfreepress.com]

  > “reported the first atmosphere ever found around a rocky planet in a star's habitable zone.”

- The exoplanet LHS 1140 b is situated approximately 50 light-years away from Earth.

  [timesfreepress.com]

  > “LHS 1140 b is one of two planets found around LHS 1140, a star about 50 light-years away.”

- Scientists detected helium around LHS 1140 b, which serves as an indicator that the planet possesses an atmosphere. [timesfreepress.com]

  > “Cherubim's team found helium around LHS 1140 b, indicating the planet has an atmosphere.”

In the image: A Falcon Heavy rocket with a payload fairing stands upright on a launchpad at twilight, showing infrastructure not mentioned in the text.

✓ 3 of 3 claims verified against sources · 1 source

timesfreepress.com

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NASA and DOE Partner for Next-Gen Space Nuclear Power

NASA and the Department of Energy signed an MoU at the White House summit to advance nuclear space technologies.

- NASA and the Department of Energy signed a Memorandum of Understanding to advance space nuclear power and propulsion. [world-nuclear-news.org]

  > "The MoU, titled 'Accelerating American Leadership in Space Nuclear Power and Propulsion', was signed on 8 October"

- NASA Administrator Jared Isaacman stated that the new era of space exploration relies on nuclear power. [world-nuclear-news.org]

  > "Nuclear power will allow us to go farther, operate longer, and field more capable spacecraft and instruments than ever before"

- The space agency aims to launch the SR-1 Freedom mission by late 2028. [world-nuclear-news.org]

  > "The space agency is targeting late 2028 to launch Space Reactor-1 Freedom (SR-1)"

- NASA is working toward deploying a lunar surface reactor by 2030 following an executive order. [world-nuclear-news.org]

  > "NASA has been directed to develop a launch-ready lunar surface reactor by 2030 under the 2025 Executive Order"

In the image: The image shows NASA Administrator Jared Isaacman and US Secretary of Energy Chris Wright on stage at the Science: A New Golden Age summit.

✓ 4 of 4 claims verified against sources · 1 source

world-nuclear-news.org

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NASA Shares Image of Stupendous 'Smiling' Galaxy Clusters

NASA has shared an amazing image captured by the Hubble Space Telescope showing galaxy clusters that resemble a smiling face.

- NASA shared an image of a galaxy cluster designated as SDSS J1038+4849 that appears to form a happy face. [lebanondemocrat.com]

  > "The cluster, designated as SDSS J1038+4849, appears to have two eyes and a nose as part of a happy face."

In the image: The image visually demonstrates the smiling face formation with glowing galaxy clusters acting as eyes and a curved cosmic arc forming a smile.

✓ 1 of 1 claims verified against sources · 2 sources

lebanondemocrat.com · myleaderpaper.com

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World's tiniest flowering plant could feed deep space astronauts

New research suggests that the world's smallest flowering plant would make an ideal food crop for future space missions.

- Researchers state that the world's smallest flowering plant would make an ideal food crop for future space missions. [lebanondemocrat.com]

  > "The world's smallest flowering plant would be an "ideal" food crop for future space missions"

In the image: A tiny duckweed plant resting on the sharpened tip of a pencil, illustrating its extremely small size relative to writing instruments.

✓ 1 of 1 claims verified against sources · 1 source

lebanondemocrat.com

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NASA Asks Commercial Businesses to Develop New Space Stations for Low Earth Orbit

As NASA prepares to decommission the aging International Space Station by the end of the decade, the agency is seeking commercial proposals for future private space stations.

- NASA has initiated the process by issuing a request for proposals from commercial space industry organizations to develop new low Earth orbit space stations. [houstonpublicmedia.org]

  > “On Friday, the space agency released its request for proposals for commercial companies to submit plans to develop commercial space stations”

- The proposal submission window remains open until December 8, with contract awards anticipated around April 15, 2027.

  [houstonpublicmedia.org]

  > “The period for proposal submissions lasts until Dec. 8. NASA anticipated it will award contracts by spring 2027, around April 15.”

- Initial contracts are planned to last roughly eight years, with potential extensions reaching a total of about fifteen years.

  [houstonpublicmedia.org]

  > “The initial recipients will receive contracts lasting about eight years, with potential extensions totaling about 15 years in length.”

- NASA selected SpaceX in 2024 to build a vehicle tasked with ultimately deorbiting the International Space Station.

  [houstonpublicmedia.org]

  > “In 2024, NASA selected SpaceX to develop a space vehicle that will deorbit the ISS”

In the image: An astronaut conducting a spacewalk outside the International Space Station with Earth's horizon in the background.

✓ 4 of 4 claims verified against sources · 1 source

houstonpublicmedia.org

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Pakistan Imposes Emergency in Khyber Pakhtunkhwa Amid Protests for Imran Khan

The federal government has suspended the provincial administration in Khyber Pakhtunkhwa and instituted governor's rule, citing security concerns and escalating political tensions surrounding jailed former Prime Minister Imran Khan.

- The federal government imposed emergency rule in Khyber Pakhtunkhwa under Articles 232 and 234 of the constitution.
  [news.webindia123.com]
  [vietnamtribune.com]
  [hindustantimes.com]
  > “The emergency rule was authorised under Articles 232 and 234 of the Pakistani Constitution following presidential approval of recommendations submitted by Prime Minister Shehbaz Sharifs federal cabinet”
- Imran Khan's son, Kasim Khan, strongly condemned the federal takeover and criticized Army Chief Asim Munir. [news.webindia123.com]
  [vietnamtribune.com]
  > “Accusing Army Chief Asim Munir of invoking 'terrorism to justify another military operation in Pashtun lands,' Kasim asserted that 'Pakistan belongs to its people, not its military.'”
- Authorities enacted stringent security crackdowns, including deploying troops, setting up container barricades, and throttling telecommunication services.
  [news.webindia123.com]
  [vietnamtribune.com]
  [hindustantimes.com]
  > “Telecom sources reported that mobile network and data services were throttled or shut down in selected urban pockets, alongside heavy police presence and shipping container barricades”
- PTI representatives and officials reported that hundreds of activists and three of Khan's sisters were detained ahead of planned protests. [hindustantimes.com]
  > “Hundreds of PTI activists and three of Khan’s sisters have been detained ahead of the planned march, in what the party described as one of the strongest crackdowns in years”

In the image: The image shows protesters, including Imran Khan's family members and supporters, marching outdoors while surrounded by heavy media and security presence.

✓ 4 of 4 claims verified against sources · 4 sources
news.webindia123.com · vietnamtribune.com · hindustantimes.com · siasat.com

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Two Pakistan Army Pilots Killed in Training Aircraft Crash

A military training aircraft crashed near Quetta due to a technical fault, resulting in the deaths of both pilots on board.

- Two Pakistan Army pilots were killed when their training aircraft crashed near Quetta. [article.wn.com]
  > “Two Pakistan Army Pilots Killed As Training Aircraft Crashes Near Quetta”
- The crash was attributed to a technical fault. [article.wn.com]
  > “The Mushshak training aircraft crashed due to a technical fault”
- The victims included an instructor and a trainee pilot. [article.wn.com]
  > “killing an instructor and a trainee pilot”

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article.wn.com

You engaged with 1 of 3 Pakistan cricket team stories you saw



Pharmacogenomics Shifts Prescribing Away From One-Size-Fits-All Approaches

Using genetic data to select medication and dosages can help prevent adverse drug reactions and protect patients from dangerous toxicities.

- Traditional medical prescribing relies on a standard method that treats diagnosed conditions with a predetermined first-line drug. [pharmacytimes.com]

  > “Modern medicine has traditionally relied on a reactive, one-size-fits-all prescribing approach.”

- Standardized drug dosing ignores biological differences among individuals, which can cause treatment failures and adverse reactions.

  [pharmacytimes.com]

  > “Unfortunately, this overlooks the biological complexity of individual patients, often resulting in treatment failures or severe adverse drug reactions.”

- Pharmacogenomic guidance leverages genetic data so patients receive optimal drug selections and doses immediately.

  [pharmacytimes.com]

> “Pharmacogenomic-guided prescribing uses genetic data to protect patients from dangerous toxicities and get them the right drug at the right dose from the start.”

In the image: The image features a floating transparent capsule filled with golden droplets alongside a molecular DNA double helix.

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pharmacytimes.com

Exploring: a Biology and medical research story outside your usual topics, to learn whether you want more



Missing Alabama Man Nathaniel Jackson Found Dead in Attalla

Human remains discovered in Attalla, Alabama, in September have been positively identified as Nathaniel Jackson, who was reported missing in July.

- Human remains were found on Bobby Austin Drive in Attalla on September 29. [wvtm13.com]

  > "The Attalla Police Department reported the human remains were found Sept. 29 on Bobby Austin Drive."

- Chief of Police Doug Jordan confirmed the remains belonged to missing man Nathaniel Jackson. [wvtm13.com]

  > "According to Chief of Police Doug Jordan, the remains were positively identified as Nathaniel Jackson."

- Nathaniel Jackson had been reported missing since July 29. [wvtm13.com]

  > "Police say Jackson was reported missing on July 29."

- The recovered remains were sent to the Alabama Department of Forensic Sciences for identification. [wvtm13.com]

  > "Upon their discovery in September, the remains were sent to the Alabama Department of Forensic Sciences for examination and identification."

In the image: The map provides a geographic overview of Attalla, Alabama, highlighting regional highways such as I-59, I-759, US-278, and US-431 surrounding the city area mentioned in the article.

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wvtm13.com

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That's today's paper. Your next edition arrives tomorrow morning.